

Exercises in Number Theory

Section A

1. Show that there is no positive integer n such that $n + k^2$ is a perfect square for at least n positive integer values of k .

(Romania 1998). Suppose there exist n positive integers $k_1 < k_2 < \dots < k_n$ such that $n + k_i^2$ is a perfect square for every $i = 1, 2, \dots, n$. Let $n + k_i^2 = m_i^2$. Then $m_1 < m_2 < \dots < m_n$ and $m_1 + k_1 < m_2 + k_2 < \dots < m_n + k_n$. But $n = (m_i + k_i)(m_i - k_i)$ for $i = 1, 2, \dots, n$, so n has at least n different divisors greater than 1, a contradiction.

Remark: By contradiction.

2. Let $n = 7 + 7^2 + 7^3 + \dots + 7^{2017}$.
(a) Show that the remainders of the divisions of 7^{2018} by 6 and by 48 are equal.
(b) Determine the last two digits of the number $6n$.

(Romania 2017). (a) Note $7n = 7^2 + 7^3 + \dots + 7^{2018}$, hence $6n = 7n - n = 7^{2018} - 7$. It follows that $7^{2018} = 6n + 7 = 6(n+1) + 1$, and the remainder of the division by 6 of the number 7^{2018} is 1. Also

$n = 7 + 7^2(1+7) + 7^4(1+7) + \dots + 7^{2016}(1+7) = 7 + 8(7^2 + 7^4 + \dots + 7^{2016}) = 7 + 8p$, where $p = 7^2 + 7^4 + \dots + 7^{2016}$. Hence

$7^{2018} = 6(8p+7) + 7 = 48p + 69 = 48(p+1) + 1$. The remainder is 1 again.

- (b) Now $6n = 7^{2018} - 7$. We check the last two digits of the number of the form 7^{4k} is 01, of 7^{4k+1} is 07, of 7^{4k+2} is 49, of 7^{4k+3} is 43. Therefore the last two digits of $7^{2018} = 7^{4 \cdot 504} + 2 = 49$, the last two digits of $6n$ is 42.

Remark: Skill.

3. Find integers n , $n \geq 3$, having the property: there exist distinct integers $a_1, a_2, \dots, a_n$, such that $a_1! a_2! \dots a_{n-1}! = a_n!$.

(Romania 2004). Indeed for $n = 3$, $3!5! = 6!$. Assume for some n , there exist $a_1 < a_2 < \dots < a_n$, such that $a_1! a_2! \dots a_{n-1}! = a_n!$. Let $b = a_n! - 1$, then $b > a$ and $a_1! a_2! \dots a_{n-1}! b! = a_n! (a_n! - 1)! = (a_n!)!$. Now take b as our new a_n and new a_{n+1} as $a_n!$.

Remark: Induction.

Section B

1. Show that there exist infinitely many pairs of positive integers (m, n) , such that $\frac{m+1}{n} + \frac{n+1}{m}$ are integers.

(British MO 2007). First check $(m, n) = (1, 2)$ satisfies the requirement. Assume now there exist integers m, n (without loss of generality assume $m < n$) such that

$\frac{m+1}{n} + \frac{n+1}{m}$ is a positive integer k . Then $k = \frac{1}{n}(\frac{n(n+1)}{m} + m + 1)$, hence

$kn = \frac{n(n+1)}{m} + m + 1$. This implies $\frac{n(n+1)}{m}$ is an integer ($= r$ say), hence

$k = \frac{1}{n}(r + m + 1) = \frac{1}{n}(r + \frac{n(n+1)}{r} + 1) = \frac{r+1}{n} + \frac{n+1}{r}$ is an integer. Hence if (m, n)

is a required pair, then so is (n, r) . But $nr > mr = n(n+1) > n^2$, hence $r > n$, which implies $n + r > m + n$, and we have a bigger (new) pair.

Remark: Use of descent.

2. Let a and b be positive integers and such that for every positive integer n ,

$(a^n + n) \mid (b^n + n)$. Show that $a = b$.

(46th IMO 2005 shortlist) Assume $a \neq b$. Take $n = 1$, get $(a+1) \mid (b+1)$, hence

$b > a$. Let p be a prime larger than b , n an integer satisfying $n \equiv 1 \pmod{p-1}$ and $n \equiv -a \pmod{p}$. This n exists because of the Chinese Remainder Theorem, for example let $n = (a+1)(p-1) + 1$. Now from Fermat's Little Theorem,

$a^n = a^{k(p-1)+1} \equiv a \pmod{p}$, hence $a^n + n \equiv a + n \equiv a - a \equiv 0 \pmod{p}$. This implies

$p \mid b^n + n$, but again by Fermat's Little Theorem $b^n + n \equiv b - a \pmod{p}$. Hence

$p \mid (b - a)$, a contradiction.

Remark: By contradiction, use of the CRT and F/T.

3. Given a positive odd integer n , show that the arithmetic mean of the fractional

parts $\left\{ \frac{k^{2n}}{p} \right\}, k = 1, 2, \dots, \frac{p-1}{2}$, is the same for infinitely many primes p .

(Romania BMO 2017 shortlist). It suffices to show that $\sum_{k=1}^{(p-1)/2} \left\{ \frac{k^{2n}}{p} \right\} = \frac{p-1}{4}$ for every prime p congruent to 1 modulo 4. Now let p be a prime congruent to 1 modulo 4, let $K = \{1, 2, \dots, (p-1)/2\}$, let $r_k, k \in K$, be the remainder k^2 modulo p , and let $K^- = \{k \mid r_k < \frac{p}{2}\}$ and $K^+ = \{k \mid r_k > \frac{p}{2}\}$. Clearly the r_k are pairwise distinct and they form the set of all quadratic residues modulo p . Since $p \equiv 1 \pmod{4}$, -1 is a residue, and hence a is a square if and only if $-a$ is a square, hence there is a bijection $f : K^- \rightarrow K^+$ such that $r_k + r_{f(k)} = p$ for all k in K^- , thus $|K^-| = |K^+| = \frac{p-1}{4}$. Now since n is odd, $k^{2n} + f(k)^{2n} \equiv 0 \pmod{p}$ for all k in K^- , so if r'_k is the remainder k^{2n} leaves upon division by p , then

$p - r'_k$ is the remainder $f(k)^{2n}$. Consequently

$$\begin{aligned} \sum_{k=1}^{(p-1)/2} \left\{ \frac{k^{2n}}{p} \right\} &= \sum_{k \in K^-} \left\{ \frac{k^{2n}}{p} \right\} + \sum_{k \in K^+} \left\{ \frac{k^{2n}}{p} \right\} = \sum_{k \in K^-} \left(\left\{ \frac{k^{2n}}{p} \right\} + \left\{ \frac{f(k)^{2n}}{p} \right\} \right) \\ &= \sum_{k \in K^-} (r'_k / p + (1 - r'_k / p)) = |K^-| = \frac{p-1}{4}. \end{aligned}$$

Remark: Understand the residue property and the quadratic reciprocity.

Section C

1. Let p be a prime and such that $p^2 \mid 2^{p-1} - 1$. Show that for any n , $(p-1)(p! + 2^n)$ has at least three distinct prime factors.

(Bulgaria 2006). Since $(p-1) \mid p!$, thus the GCD of $p-1$ and $p! + 2^n$ is a power of 2. Now we show each of $p-1$ and $p! + 2^n$ has at least one prime factors. First suppose $p-1 = 2^k$, then $p = 2^k + 1$. If $s \geq 3$ is an odd factor of k , then $p = (2^s + 1)A$, and p is not prime, impossible. Hence $k = 2^t$, thus

$$\begin{aligned} 2^{p-1} - 1 &= 2^{2^k} - 1 = (2^{2^{k-1}} - 1)(2^{2^{k-1}} + 1) = \dots \\ &= (2^{2^t} - 1)(2^{2^t} + 1) \dots (2^{2^{k-1}} - 1). \end{aligned}$$

Note that when $l > t$, $(2^{2^l} + 1, 2^{2^t} + 1) = 1$. Also $2^{2^t} - 1 < p = 2^{2^t} + 1$, thus $p^2 \nmid 2^{p-1} - 1$, a contradiction. So $p-1$ contains an odd factor.

Suppose now $p! + 2^n = 2^k$, then $k > n$, and $p! = 2^n(2^{k-n} - 1)$, then $p \mid 2^m - 1$, where $m = k - n$. If t is the smallest integer satisfying $p \mid 2^t - 1$, then t (order) divides m . By Fermat's little theorem $t \mid p-1$. Let $p-1 = lt$, then

$$2^{p-1} - 1 = (2^t - 1)(2^{t(l-1)} + 2^{t(l-2)} + \dots + 2^t + 1).$$

But $2^t \equiv 1 \pmod{p}$, hence $2^{t(l-1)} + 2^{t(l-2)} + \dots + 2^t + 1 \equiv l \not\equiv 0 \pmod{p}$. This implies $p^2 \mid 2^t - 1$, or $p^2 \mid 2^m - 1$, or $p^2 \mid p!$, contradiction.

Remark: Related to Fermat, Wieferich primes and other classical things.

2. Find all positive integers n ($n > 1$), such that there exists a unique integer a , ($0 < a \leq n!$) satisfying $a^n + 1$ is divisible by $n!$.

(2005 46th IMO Shortlist). The answer is all the primes. For $n = 2$, $a = 1$ is the only integer. For $n > 2$ and is even, a^n is a perfect square, and $a^n + 1$ is 1 or 2 modulo 4, but $n!$ is divisible by 4, impossible.

Now assume n is odd and is an odd prime p . Assuming there exists a unique a ($0 < a \leq p!$), and such that $p! \mid (a^p + 1)$. By Fermat's little theorem

$a^p + 1 \equiv a + 1 \pmod{p}$, hence $p \mid a + 1$. We now want to prove $\frac{a^p + 1}{a + 1}$ has no prime factor q which is less than p . By contradiction, suppose $q < p$ and

$q \mid \frac{a^p + 1}{a + 1}$. Now $\frac{a^p + 1}{a + 1} = \sum_{i=0}^{p-1} (-a)^i$ is odd, so is q . Thus $a^p \equiv -1 \pmod{q}$, so

$a^{2p} \equiv 1 \pmod{q}$. So a and q relatively prime, and $a^{q-1} \equiv 1 \pmod{q}$. Take $d = (2p, q-1)$, get $a^d \equiv 1 \pmod{q}$. But $q < p$, get $d = 2$. Or $a \equiv \pm 1 \pmod{q}$.

When $a \equiv 1 \pmod{q}$, then $\frac{a^p + 1}{a + 1} = \sum_{i=0}^{p-1} (-a)^i = 1 \pmod{q}$, a contradiction. When

$a \equiv -1 \pmod{q}$, then $\frac{a^p + 1}{a + 1} = \sum_{i=0}^{p-1} (-a)^i = p \pmod{q}$, or $q \mid p$, a contradiction.

Hence $\frac{a^p + 1}{a + 1}$ contains no prime factor less than p , and $(p-1)! \nmid (a+1) \frac{a^p + 1}{a + 1}$, so

$(p-1)! \nmid (a+1)$, also as $p \nmid a+1$, get $p! \nmid (a+1)$, or we may take $a = p! - 1$.

Finally if n is odd composite, take p the smallest prime factor of n and such that

$p^\alpha \parallel n!$, (just divide, note $\alpha \geq 2$). Now $2p < p^2 \leq n$, hence

$n! = 1 \times 2 \times \dots \times p \times \dots \times 2p \times \dots \times n$. Now let $m = \frac{n!}{p^\alpha}$, take any a satisfying

$a \equiv -1 \pmod{p^{\alpha-1}m}$, or $a \equiv -1 + p^{\alpha-1}mk$, then

$$a^p = (-1 + p^{\alpha-1}mk)^p = -1 + p^\alpha mk + \sum_{j=2}^p (-1)^{p-j} C_j^p p^{(\alpha-1)j} (mk)^j = -1 + p^\alpha M.$$

Here M is an integer, since for any $j \geq 2, \alpha \geq 2$, have $(\alpha-1)j \geq \alpha$. Thus

$p^\alpha \mid (a^p + 1)$, and then $p^\alpha \mid (a^n + 1)$. Also $m \mid (a+1)$, or $m \mid (a^n + 1)$. But p and m

are relatively prime, so all such a will satisfy $n! \mid (a^n + 1)$. But in the interval $[1, n!]$

there are p ($p > 2$) possible a , namely when $k = 1, 2, \dots, p$, contradicting uniqueness of a .

Remark: Complicated and Involved.

3. Find all integers n ($n > 1$) with the following property: There exist n integers $b_1, b_2, \dots, b_n$, not all equal, so that for any positive integer k , there exist integers

$a, b > 1$, satisfying $(b_1 + k)(b_2 + k) \dots (b_n + k) = a^b$.

(Russia MO 2008). Those are exactly the composite numbers $n > 1$.

Indeed if $n = rs$, (n composite) where $r > 1, s > 1$. Let

$b_1 = b_2 = \dots = b_r = 1, b_{r+1} = \dots = b_n = 2$. Then for any k ,

$$\begin{aligned} & (1+k)(1+k) \dots (1+k)(2+k)(2+k) \dots (2+k) \\ &= (1+k)^r (2+k)^{(s-1)r} = ((1+k)(2+k)^{(s-1)})^r = a^b. \end{aligned}$$

Suppose now n is a prime and there exist $b_1, b_2, \dots, b_n$, not all equal, so that for any positive integer k , there exist integers $a, b > 1$, satisfying

$(b_1 + k)(b_2 + k) \dots (b_n + k) = a^b$. We assume $b_1, b_2, \dots, b_l$ ($l > 1$) are all distinct, and

for any $b_{l+1}, b_{l+2}, \dots, b_n$, each of them is equal to one of $b_1, b_2, \dots, b_l$. Among them there are s_i of them equal b_i , $1 \leq i \leq l$, and $s_1 + s_2 + \dots + s_l = n$. Now let $p_1, p_2, \dots, p_l$ be l distinct primes, each larger than the maximum of b_i . Consider the system of equations, $b_i + m \equiv p_i \pmod{p_i^2}$, $1 \leq i \leq l$. (Its solution m is guaranteed by the Chinese remainder theorem.) This implies $b_i + m$ is a multiple of p_i but not p_i^2 . On the other hand, for $i \neq j$, where $1 \leq i, j \leq l$, as $|b_i - b_j| < p$, $b_j + m$ is not a multiple of p_i , we get $(b_1 + k)(b_2 + k) \dots (b_n + k) = a^b$ is a multiple of $p_i^{s_i}$ but not $p_i^{s_i+1}$, hence we have $b|s_i, 1 \leq i \leq l$. But $s_1 + s_2 + \dots + s_l = n$, hence $b|n$. But $l > 1$, hence $b < n$. But n is prime, hence $b = 1$, a contradiction.

Remark: precise, vigorous, curious and interesting, nice use of CRT.

27 August 2023